

Tableau Assignment – Retail Dataset Complete Step-by-Step Guide

Dataset Checked:  YES, your dataset is PERFECT for this assignment.

The dataset contains these columns:

Column	Meaning
InvoiceNo	Order/Invoice Number
StockCode	Product Code
Description	Product Name
Quantity	Number of Products Sold
InvoiceDate	Date of Purchase
UnitPrice	Price per Unit
CustomerID	Customer Identifier
Country	Customer Country

This dataset is suitable for all assignment questions.

BEFORE STARTING TABLEAU

Install Tableau Public/Desktop

Download Tableau Public:

  Tableau Public Download  <https://public.tableau.com/en-us/s/download> 

OR use Tableau Desktop if already installed.

STEP 1 — OPEN THE DATASET IN TABLEAU

1. Open Tableau

After opening Tableau:

- Click **Microsoft Excel**
- Browse and select:

- Online Retail.xlsx
-

2. Load the Sheet

You will see the Excel sheet:

- Drag **Online Retail** sheet to the center area
- Tableau will load the data preview

Now click:

→ **Sheet 1** at bottom left

IMPORTANT DATA CLEANING (VERY IMPORTANT)

Before creating charts, perform these steps.

STEP 2 — CREATE TOTAL SALES FIELD

Currently dataset has:

- Quantity
- UnitPrice

But NOT total sales.

We must create:

Total Sales = Quantity × UnitPrice

How to Create Calculated Field

Steps:

1. In left Data pane
2. Right click empty area
3. Click:
 - **Create Calculated Field**

Field Name

Type:

Total Sales

Formula

Type:

[Quantity] * [UnitPrice]

Click:

☒ OK

Now Tableau creates a new measure called:

- Total Sales
-

OPTIONAL CLEANING (RECOMMENDED)

Some records may contain:

- Negative quantity
- Cancelled invoices

To improve charts:

Apply Filter**Remove Negative Quantity**

1. Drag Quantity to Filters
2. Select:
 - At least = 1

OR simply exclude negatives.

PART (A)

Find and Plot Top 10 Products Based on Total Sale

OBJECTIVE

Find products generating highest sales.

We will use:

- Description (Product Name)
 - Total Sales
-

STEP-BY-STEP

Step 1

Open a new worksheet.

Rename sheet:

Top 10 Products

Step 2

Drag:

- Description → Rows
-

Step 3

Drag:

- Total Sales → Columns

Now Tableau creates a bar chart.

Step 4 — Sort Descending

Click sort descending icon on toolbar.

Now highest sales products appear at top.

Step 5 — Filter Top 10

Steps:

1. Right click Description
2. Click:
 - Filter
3. Open:
 - Top tab
4. Select:
 - By Field

Set:

Option Value

Top 10

By Field Sum

Field Total Sales

Click:

☒ OK

STEP 6 — Add Labels

1. Click Marks card
2. Click Label
3. Check:
 - Show Mark Labels

FINAL RESULT

You now have:

- ✓ Top 10 products by total sales.
-

PART (B)

Find and Plot Product Contribution to Total Sale

OBJECTIVE

Show percentage contribution of products in overall sales.

Best chart:

- ✓ Pie Chart

OR

- ✓ Treemap

Treemap looks more professional.

METHOD 1 — PIE CHART

Step 1

Create new sheet.

Rename:

Product Contribution

Step 2

Drag:

- Description → Color
-

Step 3

Drag:

- Total Sales → Angle
 - Total Sales → Label
-

Step 4

From Marks dropdown:

Select:

Pie

Step 5 — Convert to Percentage

1. Right click SUM(Total Sales)
 2. Quick Table Calculation
 3. Percent of Total
-

Step 6 — Show Labels

1. Click Label
 2. Check:
 - Show Mark Labels
-

BETTER METHOD — TREEMAP (Recommended)

Steps

1. New Sheet
2. Drag Description → Label
3. Drag Total Sales → Size
4. Drag Total Sales → Color

5. Click:

- Show Me

6. Select:

- Treemap

Now bigger boxes represent higher contribution.

FINAL RESULT

- ✓ Product contribution to total sales displayed.
-

PART (C)

Find and Plot Month Wise Sales in Year 2010 in Descending Order

OBJECTIVE

Display monthly sales for only year 2010.

STEP-BY-STEP

Step 1

Create new worksheet.

Rename:

2010 Monthly Sales

Step 2 — Filter Year 2010

Drag:

- InvoiceDate → Filters

A popup appears.

Select:

Years

Click:

☒ Next

Select:

2010

Click:

☒ OK

Step 3

Drag:

- InvoiceDate → Columns
-

Step 4

Click dropdown on InvoiceDate field.

Select:

MONTH(InvoiceDate)

Use discrete Month.

Step 5

Drag:

- Total Sales → Rows

Now month-wise sales chart appears.

Step 6 — Sort Descending

Click sort descending.

Now months with highest sales appear first.

Step 7 — Add Labels

1. Click Label
2. Enable:
 - Show Mark Labels

BEST CHART TYPE

Recommended:

☒ Bar Chart

Alternative:

☒ Line Chart

FINAL RESULT

☒ Month-wise sales for 2010 in descending order.

PART (D)

Find and Plot Most Loyal Customers Based on Purchase Order

OBJECTIVE

Identify customers who made maximum orders.

We use:

- CustomerID
- InvoiceNo

Loyal customer = customer with highest number of orders.

STEP-BY-STEP

Step 1

Create new worksheet.

Rename:

Loyal Customers

Step 2

Drag:

- CustomerID → Rows
-

Step 3

Drag:

- InvoiceNo → Columns
-

Step 4

Change aggregation to:

COUNTD(InvoiceNo)

COUNTD = Count Distinct Orders.

How?

1. Click dropdown on SUM(InvoiceNo)
 2. Select:
 - Measure
 - Count Distinct
-

Step 5 — Sort Descending

Click sort descending.

Now most loyal customers appear on top.

Step 6 — Filter Top Customers (Optional)

1. Right click CustomerID
 2. Filter
 3. Top
 4. Top 10 by COUNTD(InvoiceNo)
-

Step 7 — Add Labels

Enable mark labels.

FINAL RESULT

☒ Most loyal customers chart ready.

PART (E)

Find and Plot Yearly Sales Comparison

OBJECTIVE

Compare sales across different years.

STEP-BY-STEP

Step 1

Create new worksheet.

Rename:

Yearly Sales Comparison

Step 2

Drag:

- InvoiceDate → Columns
-

Step 3

Change date to:

YEAR(InvoiceDate)

Step 4

Drag:

- Total Sales → Rows
-

Step 5

Choose Chart Type

Recommended:

☒ Side-by-side bar chart

OR

☒ Line chart

Step 6 — Add Labels

Enable labels.

FINAL RESULT

☒ Year-wise sales comparison completed.

PART (F)

Find and Plot Country Wise Total Sales Price on Geospatial Graph

OBJECTIVE

Show sales by country using world map.

This is one of the most important Tableau visualizations.

STEP-BY-STEP

Step 1

Create new worksheet.

Rename:

Country Sales Map

Step 2

Double-click:

- Country

Tableau automatically generates:

☒ Map visualization

because Country is geographic data.

Step 3

Drag:

- Total Sales → Color
- Total Sales → Size

Now countries with higher sales become larger/darker.

Step 4 — Add Labels

Drag:

- Country → Label

Optional:

- Drag Total Sales → Tooltip
-

BETTER MAP SETTINGS

Change Map Style

Go to:

Map → Map Layers

Choose:

- Light
 - Normal
 - Dark
-

FINAL RESULT

 Country-wise geospatial sales map created.

BONUS — CREATE DASHBOARD (HIGHLY RECOMMENDED)

Teachers usually give extra marks for dashboard.

HOW TO CREATE DASHBOARD

Step 1

Click:

Dashboard → New Dashboard

Step 2

From left pane drag all sheets:

- Top 10 Products
- Product Contribution
- 2010 Monthly Sales
- Loyal Customers
- Yearly Sales Comparison
- Country Sales Map

onto dashboard.

Step 3

Arrange charts properly.

Suggested layout:

Top Row	Bottom Row
Top Products	Country Map
Monthly Sales	Loyal Customers
Product Contribution Yearly Comparison	

ADD DASHBOARD TITLE

Click:

Show Title

Title Example:

Retail Sales Analysis Dashboard

FORMATTING TIPS (IMPORTANT FOR MARKS)

1. Use Proper Colors

Examples:

- Blue for sales
 - Green for profit/growth
 - Red for low performance
-

2. Add Labels

Always show labels for:

- Sales values
 - Percentages
 - Customer counts
-

3. Format Currency

Steps

1. Right click Total Sales
 2. Default Properties
 3. Number Format
 4. Currency
-

4. Add Tooltips

Tooltips improve dashboard quality.

5. Rename Axis Properly

Use readable titles:

- Total Sales
- Customer Orders

- Monthly Revenue
-

EXPECTED OUTPUTS SUMMARY

Part Visualization

- A Top 10 Product Sales Bar Chart
 - B Product Contribution Pie/Treemap
 - C 2010 Monthly Sales Chart
 - D Loyal Customers Bar Chart
 - E Yearly Sales Comparison
 - F Country Wise Geospatial Sales Map
-

COMMON ERRORS & SOLUTIONS

Problem	Solution
Map not showing	Ensure Country geographic role is Country/Region
Wrong sales	Create Total Sales field correctly
Negative sales	Filter Quantity > 0
Dates not grouping	Use proper Date hierarchy
Duplicate customers	Use COUNTD(InvoiceNo)

HOW TO SAVE YOUR WORK

Save Tableau Workbook

Click:

File → Save As

Save as:

HOW TO EXPORT DASHBOARD AS PDF

Steps

1. Open Dashboard
2. File
3. Export as PDF

Useful for submission.

FINAL SUBMISSION CHECKLIST

Before submission ensure:

- ☒ Dataset loaded correctly
 - ☒ Total Sales calculated field created
 - ☒ All 6 visualizations completed
 - ☒ Dashboard created
 - ☒ Proper titles added
 - ☒ Labels visible
 - ☒ Saved workbook file
 - ☒ Exported PDF/screenshots if required
-

IMPORTANT INTERVIEW / VIVA QUESTIONS

Your teacher may ask these.

Why create calculated field?

Because sales was not directly available.

Why use COUNTD?

To count unique purchase orders.

Why use geospatial chart?

To visualize regional/country sales performance.

Why use treemap?

To compare contribution proportions visually.

CONCLUSION

Your dataset is fully suitable for the assignment.

You can successfully perform:

- ✓ Product analysis
- ✓ Sales analysis
- ✓ Customer analysis
- ✓ Time-series analysis
- ✓ Geographic analysis

using this dataset in Tableau.